

JEE MAIN REVISION MASTER: MAGNETIC PROPERTIES OF MATTER SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 73 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Magnetic Materials (Dia, Para, Ferro) & Hysteresis	43 PYQs	Materials
Very High	Magnetic Dipole Moment & Earth's Magnetism	36 PYQs	Dipoles & Terrestrial

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Magnetic Materials (Dia, Para, Ferro) & Hysteresis	$\vec{B} = \mu_0(\vec{H} + \vec{M})$, $\mu_r = 1 + \chi$, $\chi_{\text{para}} = \frac{C}{T}$, $\chi_{\text{ferro}} = \frac{C}{T - T_c}$
Magnetic Dipole Moment & Earth's Magnetism	$\vec{M} = I\vec{A} = NIA$, $\mu_B = \frac{e\hbar}{2m} \approx 9.27 \times 10^{-24} \text{ A m}$, $B_H = B \cos \delta$, $\tan \delta = \frac{B_V}{B_H}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Magnetic Materials (Dia, Para, Ferro) & Hysteresis (Very High Yield — 43 PYQs)

Core Principle: Classification of magnetic materials, Curie-Weiss law, and hysteresis loop area (energy dissipation).

■ **Caution / JEE Pitfall:** Diamagnetic materials have small negative susceptibility $\chi < 0$ that is strictly independent of temperature.

■ **Speed Trick:** Permanent magnets require high retentivity and high coercivity (e.g. Alnico). Transformer cores require high permeability and low hysteresis loss (soft iron).

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "A solenoid has a core made of material with relative permeability 400 . The magnetic field produced in the interior of solenoid is 1.0 T . T..." [Ans: B]
- [2026 Morning Shift] Q#3 (MCQ): "Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). Consider a ferromagnetic material ..." [Ans: D]
- [2025 Morning Shift] Q#1 (MCQ): "The percentage increase in magnetic field (B) when space within a current carrying solenoid is filled with magnesium (magnetic susceptibilit..." [Ans: N/A]

3.2 Magnetic Dipole Moment & Earth's Magnetism (Very High Yield — 36 PYQs)

Core Principle: Current loops as magnetic dipoles, orbital electron magnetic moment, and elements of Earth's magnetic field.

■ **Caution / JEE Pitfall:** At the magnetic equator, dip angle $\delta = 0^\circ$ ($B_V = 0$). At magnetic poles, dip angle $\delta = 90^\circ$ ($B_H = 0$).

■ **Speed Trick:** Apparent dip δ_1, δ_2 in two mutually perpendicular vertical planes: $\cot^2 \delta = \cot^2 \delta_1 + \cot^2 \delta_2$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#3 (MCQ):** "Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). Consider a ferromagnetic material ..." **[Ans: D]**
- **[2025 Evening Shift] Q#2 (MCQ):** "A magnetic dipole experiences a torque of $80\sqrt{3}$ Nm when placed in uniform magnetic field in such a way that dipole moment makes angle ..." **[Ans: N/A]**
- **[2025 Evening Shift] Q#4 (MCQ):** "Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A): Magnetic monopoles..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Evening Shift (Q#2)

A solenoid has a core made of material with relative permeability 400 . The magnetic field produced in the interior of solenoid is 1.0 T . The magnetic intensity in SI units is $\alpha \times 10^5$. The value of α is ____ . (Free space permeability $\mu_0 = 4\pi \times 10^{-7}$ SI units.)

- (A) $(25 / \pi)$
- (B) $(1 / 16\pi)$ **[Correct]**
- (C) $(1 / \pi)$
- (D) $(1 / 4\pi)$

Correct Answer: B

Solution Analysis: The relative permeability of the core is $\mu_r = 400$. The magnetic field applied is, $B = 1.0\text{ T}$. The permeability of free space is $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$. The magnetic induction B in a material is related to the magnetic intensity H by the total permeability of the medium (μ) : $B = \mu H$... (i) The total permeability μ is the product of the permeability of free space and the relative permeability of the material : $\mu = \mu_0 \mu_r$ Substituting this into the first equation : $B = \mu_0 \mu_r H \Rightarrow H = \frac{B}{\mu_0 \mu_r}$ Substituting

Benchmark Problem 2: JEE Main 2026 (Online) 23rd January Morning Shift (Q#3)

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). Consider a ferromagnetic material : Assertion (A) : The individual atoms in a ferromagnetic material possess a magnetic dipole moment and interact with one another in such a way that they spontaneously align themselves forming domains. Reason (R) : At high enough temperature, the domain structure of ferromagnetic material disintegrates. Thus, magnetization will disappear at high enough temperature known as Curie temperature. In the light of the above statements, choose the correct answer from the options given below :

- (A) (A) is true but (R) is false
- (B) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (C) (A) is false but (R) is true
- (D) Both (A) and (R) are true but (R) is not the correct explanation of (A) **[Correct]**

Correct Answer: D

Solution Analysis: Ferromagnetic materials (like iron, nickel, cobalt) are characterized by the fact that their individual atoms possess a permanent magnetic dipole moment. Due to strong quantum mechanical interactions between neighbouring atoms, these dipoles spontaneously align parallel to each other within small macroscopic regions. These aligned regions are indeed called magnetic domains. Within a domain, the magnetization is saturated, even in the absence of an external field. Hence, the Assertion (A) is True. Temperature represents thermal agitation. As the temperature of a ferromagnetic material increases